

#5 CS/SE 2340 Assignment

FALL SEMESTER 2025, Sections 002, 004, 006, 501

INSTRUCTOR: DR. ALICE WANG

Submit all of your work in a PDF file to eLearning by the due date.

Note: name your PDF files for homework submission as follows:

HW<hw#>_<FirstName>_<LastName>.PDF, e.g. HW05_Jane_Doe.PDF

Goal: The goal of this assignment is to understand and perform fixed and floating point arithmetic and to understand MIPS CPU datapath and control.

Part 1 - Fixed and Floating point arithmetic (50pts)

a) Convert the decimal number 45.3125 to an unsigned fixed-point binary representation using 6 integer bits, and 6 fractional bits. Round to the nearest representable value.

b) Convert this IEEE 754 single-precision floating-point number to decimal:

0xc30fe000

c) Convert this decimal to IEEE 754 single-precision floating-point:

2000.125

d) Perform addition of two IEEE 754 single-precision floating-point numbers:

0x44fa0000 + 0x41a00000

e) Perform multiplication of two IEEE 754 single-precision floating-point numbers:

0x44fa0000 x 0x41a00000

For (d) and (e), check your math by converting both floats to binary scientific notation, performing the subtraction/multiplication, and converting the result back to IEEE 754 format. You will lose points if you convert to decimal to perform the subtraction/multiplication.

Part 2 - CPU Datapath and Control (50pts)

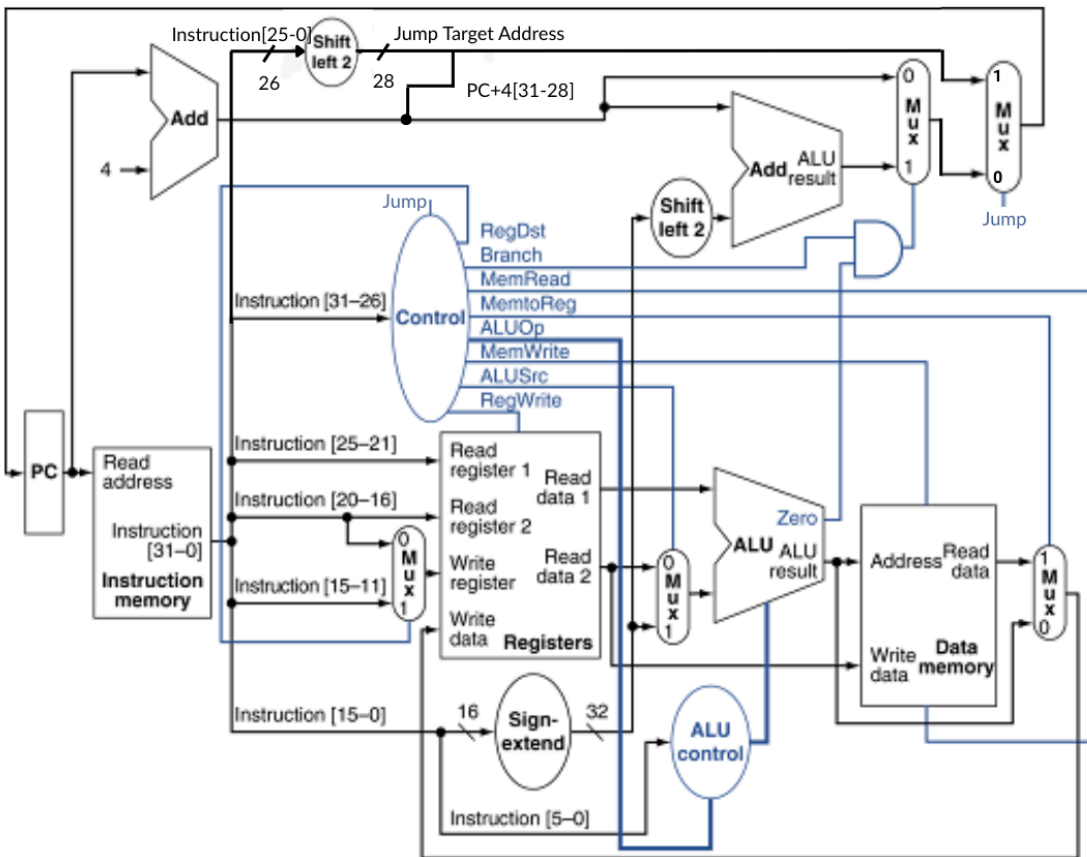
In the single-cycle MIPS processor datapath and control, highlight the datapath for the jump and link instruction (**jal**) and the (**andi**) instruction.

Also fill out the instruction format and the control signal table below.

Instruction	jal Loopcode	andi \$t0, \$t1, 4
Format type (R,I,J)		
Control signals		
Jump		
RegDst		
Branch		
MemRead		
MemtoReg		
ALU control		
MemWrite		
ALUSrc		
RegWrite		

opcode	ALUOp	Operation	funct	ALU function	ALU control
lw	00	load word	XXXXXX	add	0010
sw	00	store word	XXXXXX	add	0010
beq	01	branch equal	XXXXXX	subtract	0110
R-type	10	add	100000	add	0010
		subtract	100010	subtract	0110
		AND	100100	AND	0000
		OR	100101	OR	0001
		set-on-less-than	101010	set-on-less-than	0111

Highlight jal instruction



Highlight and instruction

